

Inheritance tax avoidance through the family firm

Isabel Micó-Millán
Banco de España ¹

Simposio de la Asociación de la Economía Española, Palma de Mallorca

December 16, 2024

¹ *The views expressed here are mine and do not necessarily represent the views of the ECB or the Bank of Spain*

Motivation

Recent wealth inequality trends have renewed interest in inheritance taxation; ongoing debate on the design of the tax ([OECD, 2021](#))

- ▷ In many countries, the design of this tax includes preferential tax treatment of **certain assets** (main residence, closely-held firms, and other)
 - ▶ Liquidity concerns ([Loutzenhiser and Mann, 2021](#); [Scheuer and Slemrod, 2021](#))
 - ▶ **Strong incentives** to engage in asset reallocation strategies to ↓ tax liabilities of heirs
- ▷ **Missing:** Evidence on inheritance tax avoidance through asset-shifting
 - ▶ How big is it? What type of asset vehicles are used? What are the fiscal costs?

Motivation

Recent wealth inequality trends have renewed interest in inheritance taxation; ongoing debate on the design of the tax ([OECD, 2021](#))

- ▷ In many countries, the design of this tax includes preferential tax treatment of **certain assets** (main residence, closely-held firms, and other)
 - ▶ Liquidity concerns ([Loutzenhiser and Mann, 2021](#); [Scheuer and Slemrod, 2021](#))
 - ▶ **Strong incentives** to engage in asset reallocation strategies to ↓ tax liabilities of heirs
- ▷ **Missing:** Evidence on inheritance tax avoidance through asset-shifting
 - ▶ How big is it? What type of asset vehicles are used? What are the fiscal costs?

Motivation

Recent wealth inequality trends have renewed interest in inheritance taxation; ongoing debate on the design of the tax ([OECD, 2021](#))

- ▷ In many countries, the design of this tax includes preferential tax treatment of **certain assets** (main residence, closely-held firms, and other)
 - ▶ Liquidity concerns ([Loutzenhiser and Mann, 2021](#); [Scheuer and Slemrod, 2021](#))
 - ▶ **Strong incentives** to engage in asset reallocation strategies to ↓ tax liabilities of heirs
- ▷ **Missing:** Evidence on inheritance tax avoidance through asset-shifting
 - ▶ How big is it? What type of asset vehicles are used? What are the fiscal costs?

This paper

This paper: Study inheritance tax avoidance through asset-shifting

Context: Inheritance tax reform in Catalonia; increase in the tax differential between assets categories for top 2% heirs

Data: Universe of inheritance tax returns; wealth tax returns

Empirical setting: Compare wealthy direct descendants (treated) to wealthy spouses and distant heirs (control)

Main outcomes:

- ▶ Estimate changes in the asset composition of inheritances & study the channels
- ▶ Tax revenue analysis

This paper

This paper: Study inheritance tax avoidance through asset-shifting

Context: Inheritance tax reform in Catalonia; increase in the tax differential between assets categories for top 2% heirs

Data: Universe of inheritance tax returns; wealth tax returns

Empirical setting: Compare wealthy direct descendants (treated) to wealthy spouses and distant heirs (control)

Main outcomes:

- ▶ Estimate changes in the asset composition of inheritances & study the channels
- ▶ Tax revenue analysis

This paper

This paper: Study inheritance tax avoidance through asset-shifting

Context: Inheritance tax reform in Catalonia; increase in the tax differential between assets categories for top 2% heirs

Data: Universe of inheritance tax returns; wealth tax returns

Empirical setting: Compare wealthy direct descendants (treated) to wealthy spouses and distant heirs (control)

Main outcomes:

- ▶ Estimate changes in the asset composition of inheritances & study the channels
- ▶ Tax revenue analysis

This paper

This paper: Study inheritance tax avoidance through asset-shifting

Context: Inheritance tax reform in Catalonia; increase in the tax differential between assets categories for top 2% heirs

Data: Universe of inheritance tax returns; wealth tax returns

Empirical setting: Compare wealthy direct descendants (treated) to wealthy spouses and distant heirs (control)

Main outcomes:

- ▶ Estimate changes in the asset composition of inheritances & study the channels
- ▶ Tax revenue analysis

This paper

This paper: Study inheritance tax avoidance through asset-shifting

Context: Inheritance tax reform in Catalonia; increase in the tax differential between assets categories for top 2% heirs

Data: Universe of inheritance tax returns; wealth tax returns

Empirical setting: Compare wealthy direct descendants (treated) to wealthy spouses and distant heirs (control)

Main outcomes:

- ▶ Estimate changes in the asset composition of inheritances & study the channels
- ▶ Tax revenue analysis

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**
 - Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Preview of the results

- 1. Composition of inheritances.** The fraction of tax-favored assets inherited by top 2% descendants $\uparrow$ **5.8 p.p**
 - ▶ **Who?** Effects entirely explained by **top 0.5% heirs** ($\uparrow$ 30 p.p)
 - ▶ **How?** Effects entirely driven by $\uparrow$ **equity in family firms**

→ Wealthy testators add private assets to their firms as capital contributions before death
- 2. Other margins of response.** No change in reported inheritances; small $\uparrow$ inter-vivos transfers; other testament planning strategies
- 3. Tax revenue analysis.** The reclassification of private wealth as business wealth $\approx$ 117M in forgone tax revenue (**27% inher. tax revenues**)

Institutional Setting

The Spanish Inheritance and Gift Tax

Institutional Setting

Tax design. Regulated at the national level²

- ▶ Progressive tax schedule (7.65% - 34%)
- ▶ Exemption thresholds; kinship bw the donor and the recipient
- ▶ Asset-specific tax benefits

Tax formula

Assessment rules and tax enforcement.

- ▶ Self-reported values backed with third-party information (bank statements, real estate registry, etc.)

Tax decentralization

- ▶ Regions can modify exemption thresholds, tax schedule & tax benefits
- ▶ Tax returns are filed in the [region of residence](#) of the deceased

²Except for Basque Country and Navarre

The Spanish Inheritance and Gift Tax

Institutional Setting

Tax design. Regulated at the national level²

- ▶ Progressive tax schedule (7.65% - 34%)
- ▶ Exemption thresholds; kinship bw the donor and the recipient
- ▶ Asset-specific tax benefits

Tax formula

Assessment rules and tax enforcement.

- ▶ Self-reported values backed with third-party information (bank statements, real estate registry, etc.)

Tax decentralization

- ▶ Regions can modify exemption thresholds, tax schedule & tax benefits
- ▶ Tax returns are filed in the **region of residence** of the deceased

²Except for Basque Country and Navarre

The Spanish Inheritance and Gift Tax

Institutional Setting

Tax design. Regulated at the national level²

- ▶ Progressive tax schedule (7.65% - 34%)
- ▶ Exemption thresholds; kinship bw the donor and the recipient
- ▶ Asset-specific tax benefits

Tax formula

Assessment rules and tax enforcement.

- ▶ Self-reported values backed with third-party information (bank statements, real estate registry, etc.)

Tax decentralization

- ▶ Regions can modify exemption thresholds, tax schedule & tax benefits
- ▶ Tax returns are filed in the [region of residence](#) of the deceased

²Except for Basque Country and Navarre

Asset-specific Inheritance Tax Benefits

Tax-favoured assets

1. Family-firm assets, agricultural land & cultural assets: 95% tax credit
2. Main residence and life insurance: capped 95-100% tax credit

Non tax-favoured assets

Other real estate property, financial wealth, other assets

Asset-specific Inheritance Tax Benefits

Tax-favoured assets

1. Family-firm assets, agricultural land & cultural assets: **95% tax credit**
→ Only 5% of asset value enters the net tax base
2. Main residence and life insurance: **capped 95-100% tax credit**
→ Only 5% of asset value enters the net tax base; limited deductible amount

Non tax-favoured assets

Other real estate property, financial wealth, other assets

Institutional Setting

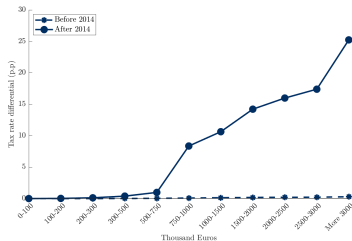
2014 Tax Reform in Catalonia

- Between 2011 and 2014, inheritance tax for close heirs was virtually 0
 - ▶ In 2011, close heirs were granted a 99% tax credit applicable to final tax liabilities
- In February 2014, the Catalan government modified the inheritance tax code
 - ▶ Decreased the exemption thresholds
 - ▶ Reduced general tax credits only for direct descendants

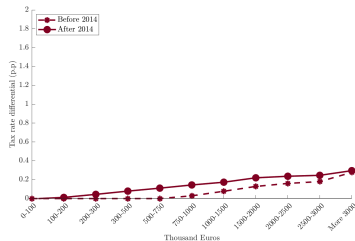
Go

Institutional Setting

2014 Inheritance Tax Reform in Catalonia



(a) Direct Descendants

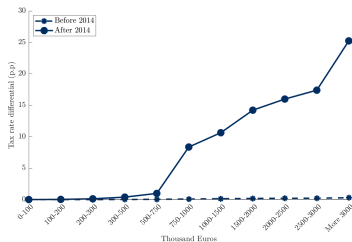


(b) Spouse

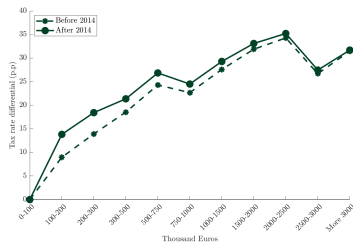
Figure: Average Tax Rate Differential bw Asset Groups - Before and After the 2014 reform

Institutional Setting

2014 Inheritance Tax Reform in Catalonia



(a) Direct Descendants



(b) Distant heirs

Figure: Average Tax Rate Differential bw Asset Groups - Before and After the 2014 reform

Data

Data

Administrative tax data

Data. Universe of inheritance and gift tax returns filed in Catalonia 2011-2019 (**Catalan Tax Agency**) Summary Statistics

- ▶ *Zoom in* on the top of the inheritance distribution
- ▶ Detailed data; link heirs sharing the same estate
- ▶ Combined with self-constructed tax simulator ([Micó-Millán, 2023](#))

Treatment and control groups. Restrict the sample to estates entirely transmitted to either descendants, spouses or distant heirs

- ▶ **Treatment group.** Direct descendants ($> 750,000$ euros)
- ▶ **Control group.** Surviving spouses & distant heirs ($> 750,000$ euros)

Summary Statistics

Treatment and Control Groups

	(1) Treated Group	(2) Control Group
	Before 2014	Before 2014
Tax Base (1000 euros)	3198.12	2519.98
Tax-favored Assets (% Tax Base)	22.03	22.22
Business Assets (% Tax Base)	11.31	9.84
Primary Residence (% Tax Base)	6.92	6.88
Life Insurance (% Tax Base)	3.19	5.42
Other Tax-favored Assets (% Tax Base)	0.82	0.08
Inheritance Tax Rate (%)	0.08	6.17
Observations	9,784	4,990

Table: Summary Statistics - Treatment and Control Group Pre-reform Period

More

Sample

Empirical Model and Results

Empirical Strategy

Diff-and-Diff Strategy

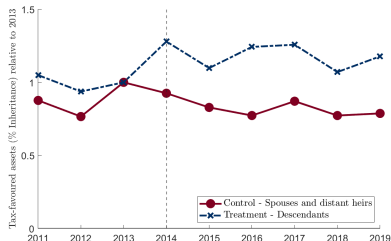
$$Y_{it} = \sum_{\substack{j=2011 \\ j \neq 2013}}^{2019} \beta_j D_{j=t} \times T_i + \gamma_t + \epsilon_{it} \quad (1)$$

- ▷ Y_{it} : outcome variable
- ▷ D_j : time dummies
- ▷ T_i : Treatment indicator (= 1 descendants; = 0 other heirs)
- ▷ λ_t year FE; ϵ_{it} the error term.

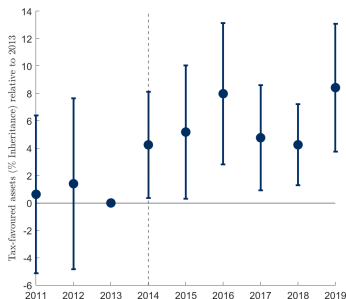
1. Asset-shifting responses

Results

Diff-and-Diff Estimates - Asset-shifting responses



(a) Treatment and control



(b) DD Estimates

Figure: % Tax-favored Assets (Total Inheritance)

Average post-treatment Δ 5.8 p.p; $\downarrow$ tax rates 1.3 p.p (41,000 Euro)

Placebo

Control groups

Wealth tax

The mechanism

Understanding asset-shifting responses

① Heterogeneity by wealth bracket

- ▶ We should expect stronger responses for larger inheritances; tax incentives are more salient Go

② The anatomy of the asset shifting

- ▶ Portfolio reallocation is not a frictionless process (Advani and Tarrant, 2021)

The mechanism

Heterogeneity by Wealth Bracket

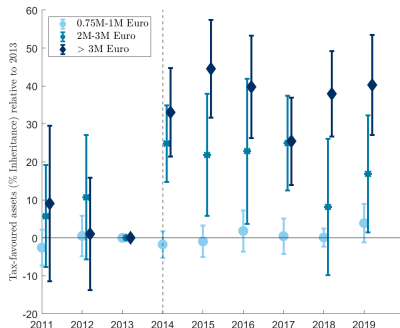


Figure: Tax-favored assets (% Total Inheritance) - By wealth bracket

The average response is entirely driven by **top 0.5% heirs (> €2M)**

Average post-treatment Δ 30 p.p; $\downarrow$ tax rates 3.8 p.p (300,000 Euro)

The mechanism

Heterogeneity by Wealth Bracket

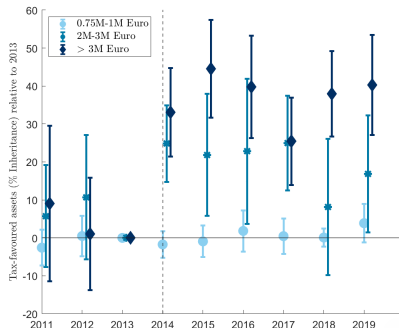


Figure: Tax-favored assets (% Total Inheritance) - By wealth bracket

The average response is entirely driven by **top 0.5% heirs** ($> \text{€}2\text{M}$)

Average post-treatment Δ 30 p.p; $\downarrow$ tax rates 3.8 p.p (300,000 Euro)

The mechanism

Understanding asset-shifting responses

① Heterogeneity by wealth bracket

- ▶ Stronger responses for larger inheritances; tax incentives should be more salient

② The anatomy of the asset shifting

- ▶ Portfolio reallocation is not a frictionless process ([Advani and Tarrant, 2021](#))

The mechanism

The Anatomy of The Asset-Shifting

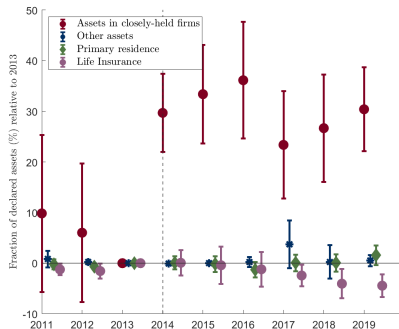


Figure: Tax-favored asset (% Total Inheritance) - By type of asset

↑ % tax-favored assets inherited $\approx$ ↑ **assets in closely held firms**

Wealthy individuals reclassify private wealth through their firms before death

Evidence WT

The mechanism

The Anatomy of The Asset-Shifting

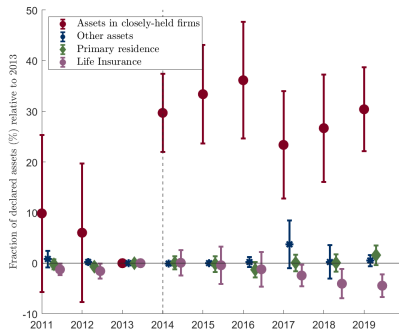


Figure: Tax-favored asset (% Total Inheritance) - By type of asset

↑ % tax-favored assets inherited $\approx$ ↑ **assets in closely held firms**

Wealthy individuals reclassify private wealth through their firms before death

Evidence WT

Family firms as a tax avoidance vehicle

The relabeling of private wealth into business wealth before death. How does this work?

1. Add assets to already-established firms
2. Set up a family firm

Family firms as a tax avoidance vehicle

The relabeling of private wealth into business wealth before death. How does this work?

1. Add assets to already-established firms
2. Set up a family firm

Family firms as a tax avoidance vehicle

The relabeling of private wealth into business wealth before death. How does this work?

1. **Add assets to already-established firms**
2. Set up a family firm

Family firms as a tax avoidance vehicles

The relabeling of private wealth into business wealth before death. How does this work?

1. Add assets to already-established firms

- (i) Inject assets as capital contributions; gives rise to new equity for shareholders
- (ii) Add certain assets as operative capital (buildings, vehicles, etc.) under some conditions
- (iii) Sell assets to the firm

2. Set up a family firm

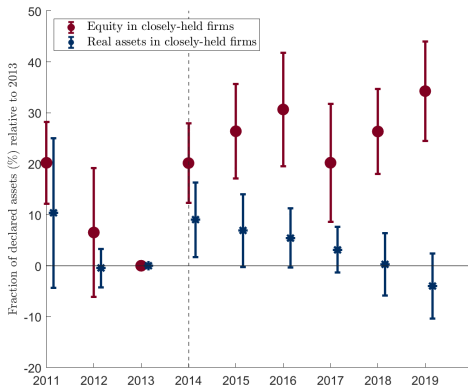
Family firms as a tax avoidance vehicles

The relabeling of private wealth into business wealth before death. How does this work?

- (i) Add assets to firms as capital contributions; implies new equity for shareholders
- (ii) Add certain assets as operative capital (buildings, vehicles, etc.) under some conditions
- (iii) Sell assets to the firm

Disentangle (i) and (ii) by studying the **type of business assets inherited** - equity vs real assets

Family firms as a tax avoidance vehicles



→ Effects work mostly through **equity in family firms**; capital contributions channel

Estimates comparable to the literature ([Duran-Cabr  et al., 2019](#); [Alvaredo and Saez, 2009](#))

3. Other margins of response

Other Margins of Response

1. Reported inheritances

Go

- ▶ No change in reported taxable inheritances ([Montserrat, 2019](#))

2. Testament planning of wealthy families

Stats

Results

- ▶ Wealthy families (i) redistribute wealth from descendants to spouses (ii) redistribute business assets from spouses to descendants **to reduce tax liabilities**

3. Inter-vivos gifts

Tax design

Admin data

- ▶ **Small positive effect** on *inter-vivos* transfers of real estate/financial wealth

Go

4. Changes in fiscal residence

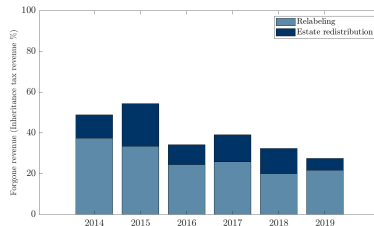
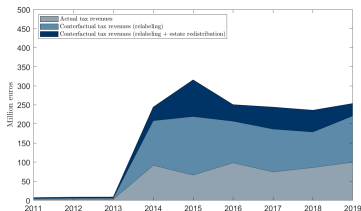
- ▶ Unlike the wealth tax ([Agrawal et al., 2023](#)), hardly significant in the Spanish setting (frequency of tax changes is high, def. region of residence, etc.)

Tax revenue implications of tax avoidance

Elasticity estimates & tax revenue analysis

- ▷ Elasticity estimates wrt tax differential ([Herry, 2023](#); [Bergolo et al., 2022](#); [Waseem, 2018](#)) Go
 - ▶ Pure asset-shifting $\epsilon \approx 17$ ([Alvaredo and Saez, 2009](#); [Duran-Cabr  et al., 2019](#))
- ▷ **Tax revenue analysis.** Use of inheritance tax simulator to carry out tax revenue counterfactuals
 1. Inheritance tax revenue without **wealth relabeling**
 2. Inheritance tax revenue without **wealth relabeling + testament planning**

Tax revenue analysis



(a) Tax revenue simulation - Heirs > €2M

(b) Forgone tax revenue (% total revenue)

- Catalonia forgoes around **27%** of inheritance tax revenues (117M) due the **reclassification of private wealth through closely held firms**

Conclusion and Policy Implications

Conclusion

- ▶ Wealthy individuals respond strongly to inheritance taxes by reallocating their wealth towards tax-exempted assets
- ▶ Their preferred and unique asset vehicle are family firms
- ▶ This entails large tax revenue losses

Policy implications

- ▶ Family-firm tax exemptions are widespread in many countries [OECD \(2021\)](#)
- ▶ Should tax design take this into account? Equity-efficiency trade-off
 - ★ Mixed evidence on liquidity effects of taxing firm succession ([Brunetti, 2006](#); [Tsoutsoura, 2015](#); [Holtz-Eakin et al., 2001](#))
 - ★ Wealthy individuals are using their firms as tax avoidance instruments; challenges progressivity of tax system

Thank you!

Back-up Slides

Spanish Inheritance Tax

Asset-specific tax benefits in Catalonia

[Back](#)

- For tax bracket $j \in [t_j^{lb}, t_j^{ub}]$

$$\text{Net Tax Base}^i = \min \left\{ 0, \left(\sum_r (\text{Tax-favoured Assets}_r - k_r) \times (1 - \text{TC}^{r,i}) + \sum_s \text{Non tax-favoured Assets}_s \right) - \text{TD}^i \right\} \quad i \in \{\text{heir, donee}\}$$

$$\text{Tax Quota}_{jt}^i = (q_j + (\text{Net Tax Base}_{jrt} - t_j^{lb}) \times \tau_{jt}) \times (1 - \text{TC}_t^i) \times \text{SF}_t$$

- **TC**: tax credit
- **TD**: tax deduction
- τ : marginal tax rate
- **SF**: pre-inheritance wealth scaling factor

- Effective tax rate:

$$\tau_{jrt}^{E,i} = \frac{\text{Tax Quota}_{jrt}^i}{\text{Net Tax Base}_{jrt}^i} \quad i \in \{\text{inher, gift}\}$$

Institutional Setting

2014 Tax Reform in Catalonia - Main changes

	Jan 2011 - Jan 2014	Feb 2014- Dec 2019		
Exemption thresholds				
<i>Descendants</i>				
Son or daughter > 21	275k	100k		
Other direct descendants	150k	50k		
<i>Spouse</i>	500k	100k		
Tax refunds				
	No tax-favored assets ³	At least one tax-favored assets	No tax-favored assets	At least one tax-favored assets
All descendants	99%	49.5%	99-20%	49.5-10%
Spouse	99%	49.5%	99%	49.5%

[Back](#)

³Excluding primary residence and life insurance

Data

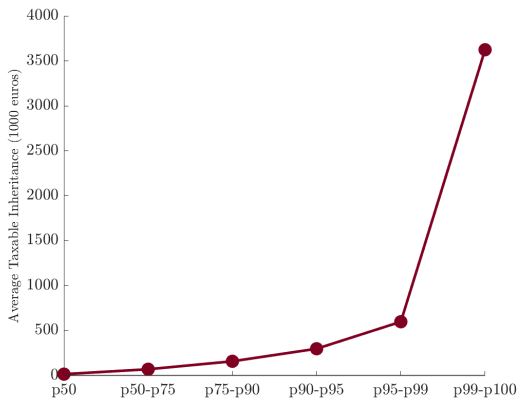
Summary Statistics - Kinship

<i>Close heirs. Group 1</i>	
Descendants < 21 age	2.3%
<i>Close heirs. Group 2</i>	
Son or daughter > 21 age	57.7%
Spouse	19.9%
Other direct descendants	5.8%
Ascendants	0.9%
<i>Distant heirs. Groups 3 & 4</i>	13.3%

Table: Summary Statistics - Heirs by Kinship

Data

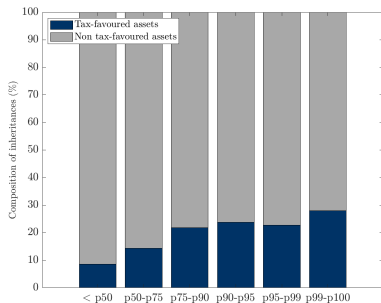
Summary Statistics - Close heirs (descendants and spouses)



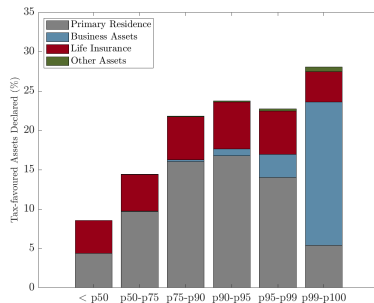
Back

Data

Summary Statistics - Close heirs



(a) Asset Composition



(b) Tax-favoured Assets Composition

Data

Summary Statistics- Treatment and Control Group

	(1) Treated Group	All	(2) Control Group Surviving spouse	Distant heirs
	Before 2014		Before 2014	
Tax Base (1000 euros)	3198.12	2519.98	1826.61	3346.90
Tax-favored Assets (% Tax Base)	22.03	22.22	21.29	23.33
Business Assets (% Tax Base)	11.31	9.84	3.79	17.07
Primary Residence (% Tax Base)	6.92	6.88	11.97	0.82
Life Insurance (% Tax Base)	3.19	5.42	5.58	5.21
Other Tax-favored Assets (% Tax Base)	0.82	0.08	0.02	0.16
Inheritance Tax Rate (%)	0.08	6.17	0.05	14.67
Observations	3433	1655	901	754
Total number of observations	9,784	4,990	2,918	2,072

Table: Summary Statistics - Treatment and Control Group Pre-reform Period

Data

Summary Statistics- Treatment and Control Group

	(1) Treated Group	(2) Control Group All
	Before 2014	Before 2014
Tax Base (1000 euros)	9545.48	8129.51
Tax-favored Assets (% Tax Base)	30.62	25.23
Business Assets (% Tax Base)	26.73	18.87
Primary Residence (% Tax Base)	2.52	3.26
Life Insurance (% Tax Base)	1.32	3.09
Other Tax-favored Assets (% Tax Base)	0.46	0.00
Inheritance Tax Rate (%)	0.10	5.86
Observations	850	337
Total number of observations	2,004	1,031

Table: Summary Statistics - Treatment and Control Group Pre-reform Period (Tax base > 2M)

Data

Sample restriction

	Descendants	Spouses	Distant heirs
Sharing estate	26.56%	46.42%	0.97%
Not sharing estate	73.44%	53.58%	99.03%

Table: Summary Statistics - Heirs with tax base > €750,000

Back

Results

Evidence from wealth tax returns

Observing the wealth portfolio of treated heirs' ascendants before death would allow to provide direct evidence on asset reallocation strategies ([Kopczuk, 2007](#))

→ Use the universe of wealth tax returns > 65 years old in Catalonia

- ▶ Comparable sample with treated heirs - Top 25% wealth taxpayers Stats
- ▶ Wealth tax-exempted assets - Main residence and family-firm assets

Results

Evidence from wealth tax returns

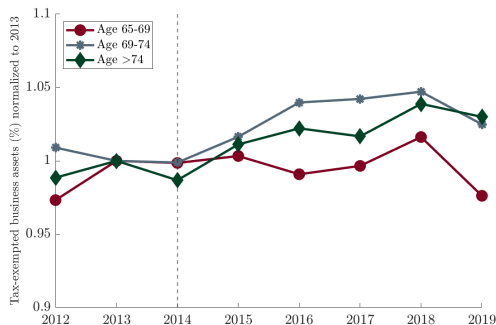


Figure: Tax-exempted Assets (% Total wealth) - Above 65 years old

Data

Comparable Wealth and Inheritance Tax Samples

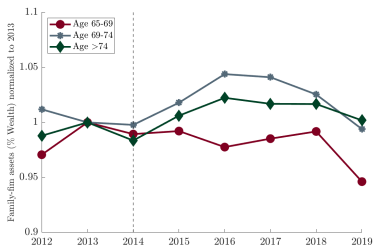
	(1) Wealth tax returns			(1) Inheritance tax returns	
	Top 25%	Top 10%	Top 5%	Top 2%	Top 0.5%
Total wealth/estate declared	7,682	15258.65	25612.71	7,544	24365.78
Family-firm assets (% Total wealth)	28.58	43.92	52.14	10.83	24.50
Main dwelling (% Total wealth)	5.84	3.52	2.51	6.91	2.72

Table: Summary Statistics - Wealth Tax Returns vs Inheritance Tax Returns

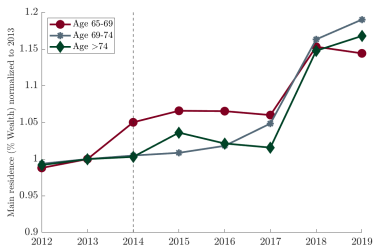
Back

Mechanism

Evidence from wealth portfolio composition - Taxpayers >65



(a) Family-firm Assets

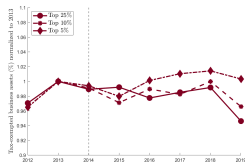


(b) Main Residence

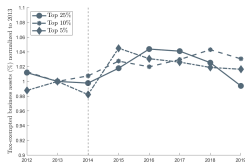
Figure: Composition of wealth portfolio - Wealth taxpayers by age group

Mechanism

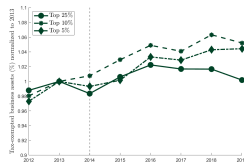
% Family-firm assets - Wealth taxpayers > 65



(a) Age 65 - 69



(b) Age 69 - 74



(c) Age > 74

Figure: Family-firm assets (% Total Wealth) - By age and wealth percentile

Back

Spanish Gift Tax

Tax design - Close heirs

Kinship tax deductions	
Exemption thresholds	-
Asset-specific tax refunds	
Equity shares in closely-held businesses	95% tax refund
Business-related real assets	95% tax refund
Agricultural	95% tax refund
Cultural property	95% tax refund
Cash for acquisition of first residence	95% tax refund; limit 125k
House for first residence	95% tax refund; limit 125k
Tax schedule	
As inheritance tax	

Table: Gift Tax - 2010-2019

Gift tax

Catalan Admin Data

- Universe of anonymized gift tax returns between 2011-2019
 - ▶ **651-form**: donor's tax base, kinship relationship, tax deductions, year and month
 - ▶ Information on pre-gift wealth of recipients; allows to identify wealthy descendants

Back

Inter-vivos gifts

Real estate and financial wealth

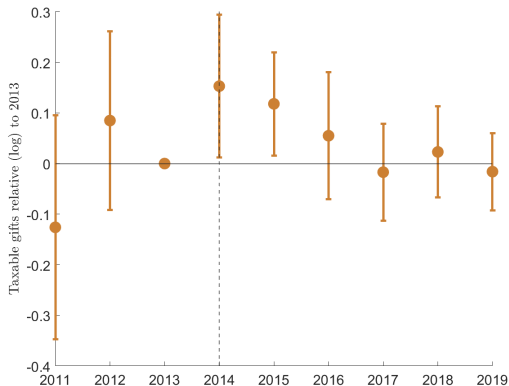
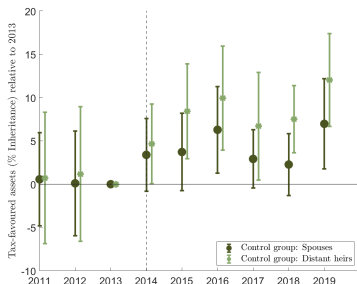


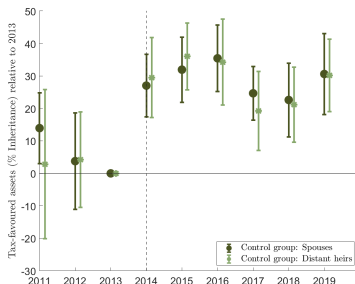
Figure: Inter-vivos transfers of real estate/financial wealth > 100K

Robustness

Choice Control Group



(a) Heirs > €750,000



(b) Heirs > €2M

Figure: Robustness- Choice of Control Group

Robustness

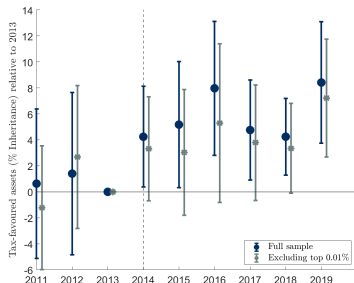
Placebo Exercise



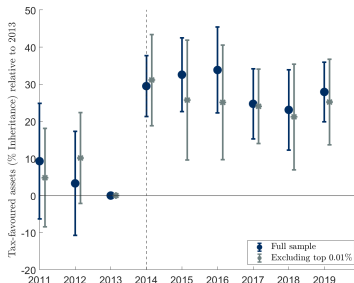
Figure: Robustness- Choice of Control Group

Robustness

Outliers



(a) Heirs > €750,000

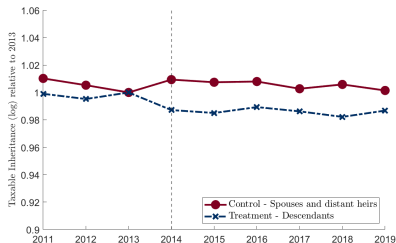


(b) Heirs > €2M

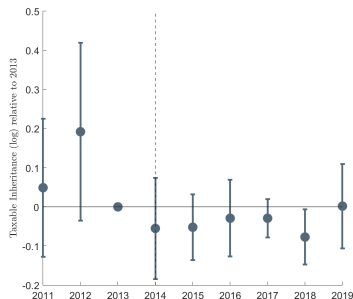
Figure: Robustness - Outliers

Results

Other Margins of Response



(a) Treatment and control



(b) DD Estimates

Figure: Inheritance tax base (log)

Other Margins of Response

Testament planning of wealthy families

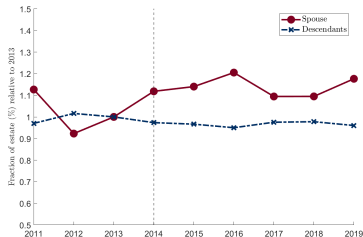
	Descendants	Spouses	Distant heirs
Sharing estate	26.56%	46.42%	0.97%
Not sharing estate	73.44%	53.58%	99.03%

Table: Summary Statistics - Heirs with tax base > €750,000

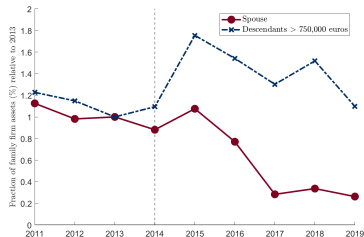
Back

Other Margins of Response

Testament planning of wealthy families



(a) % Estate inherited



(b) % Family-firm assets inherited

Figure: Testament planning of wealthy families

Inheritance tax elasticities

Framework

Conceptual Framework

Build on ([Bergolo et al., 2022](#); [Waseem, 2018](#); [Piketty and Saez, 2013](#))

$$\frac{\partial B}{\partial(\tau_{NF} - \tau_F)} = \frac{1}{(\tau_{NF} - \tau_F)} \left[-\tau_{NF} \cdot \eta_s \cdot I^{NF} + \tau_F \cdot \nu_s \cdot I^F \right]$$

- ★ η_s : total intensive margin elasticity (shifting + evasion responses)
- ★ ν_s : pure asset-shifting elasticity

In general, $\eta_s \geq \nu_s$

Estimation of elasticities (2SLS-IV)

$$\log(I_{it}^{NF}) = \eta_s(\tau_{it}^{NF} - \tau_{it}^F) + \gamma_t + u_{it}$$

$$\log(I_{it}^F) = \nu_s(\tau_{it}^{NF} - \tau_{it}^F) + \gamma_t + u_{it}$$

Instrument: $Post_t \times Treat_i$

Elasticity Estimates

	(1)	(2)
	Tax-favoured assets	Tax-favoured assets
	All heirs	Heirs with tax base > 2 million
Elasticity w.r.t $(\tau^{NF} - \tau^F)$	16.919*** (3.010)	20.228*** (3.888)
F-stat First stage	129.68	129.68
Time FE	Yes	Yes
Observations	7721	1628

Table: Pure Asset-shifting Elasticity Estimates

Pure asset-shifting elasticity similar to previous estimates for wealth tax ([Alvaredo and Saez \(2009\)](#) find $\epsilon \approx 25$; [Duran-Cabr  et al. \(2019\)](#))

Back

References I

- Advani, A. and Tarrant, H. (2021). Behavioural responses to a wealth tax. *Fiscal Studies*, 42(3-4):509–537.
- Agrawal, D. R., Foremny, D., and Martínez-Toledano, C. (2023). Wealth tax mobility and tax coordination. *American Economic Journal: Applied Economics*, Forthcoming.
- Alvaredo, F. and Saez, E. (2009). Income and wealth concentration in Spain from a historical and fiscal perspective. *Journal of the European Economic Association*, 7(5):1140–1167.
- Bergolo, M., Burdin, G., De Rosa, M., Giacobasso, M., Leites, M., and Rueda, H. (2022). How do top earners respond to taxation? Evidence from a tax reform in Uruguay. *Working paper*.
- Brunetti, M. J. (2006). The estate tax and the demise of the family business. *Journal of Public Economics*, 90(10-11):1975–1993.
- Duran-Cabré, J. M., Esteller-Moré, A., and Mas-Montserrat, M. (2019). Behavioural responses to the (re) introduction of wealth taxes: Evidence from Spain. *IEB Working Paper*.
- Herry, S. L. G. (2023). Wealth taxation and portfolio allocation. *Working paper*.

References II

- Holtz-Eakin, D., Phillips, J. W., and Rosen, H. S. (2001). Estate taxes, life insurance, and small business. *Review of Economics and Statistics*, 83(1):52–63.
- Kopczuk, W. (2007). Bequest and tax planning: Evidence from estate tax returns. *The Quarterly Journal of Economics*, 122(4):1801–1854.
- Loutzenhiser, G. and Mann, E. (2021). Liquidity issues: Solutions for the asset rich, cash poor. *Fiscal Studies*, 42(3-4):651–675.
- Micó-Millán, I. (2023). The effects of inheritance and gift taxation on upward wealth mobility at the bottom: Lessons from Spain. *Working paper*.
- Montserrat, M. M. (2019). What happens when dying gets cheaper? Behavioural responses to inheritance taxation. *Working paper*.
- OECD (2021). *Inheritance Taxation in OECD Countries*.
- Piketty, T. and Saez, E. (2013). Optimal Labor Income Taxation. In *Handbook of Public Economics*, chapter 7, pages 391–474. Elsevier.
- Scheuer, F. and Slemrod, J. (2021). Taxing our wealth. *Journal of Economic Perspectives*, 35(1):207–230.
- Tsoutsoura, M. (2015). The effect of succession taxes on family firm investment: Evidence from a natural experiment. *The Journal of Finance*, 70(2):649–688.

References III

Waseem, M. (2018). Taxes, informality and income shifting: Evidence from a recent Pakistani tax reform. *Journal of Public Economics*, 157:41–77.